Questionnaire Design

Block 8

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Agenda

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Objectives

After this chapter, you should be able to:

1. explain the purpose of a questionnaire and its objectives of asking questions that participants **can** and **will** answer, engaging participants and minimising response error;
2. understand the **trade-offs** that have to be made when designing a questionnaire;
3. describe the **process** of designing a questionnaire, the steps involved and guidelines that must be followed at each step;
4. discuss the considerations involved in designing questionnaires that work across **multiple countries and cultures**;
5. appreciate how **technology** is shaping the ways in which questionnaires are designed, delivered and experienced.

Overview

This chapter discusses the importance of questionnaire design and how researchers must put themselves *'in the shoes'* of target participants in order to create an effective questionnaire.

- There are no scientific principles that guarantee an optimal questionnaire – it is more of a **craft** honed through experience.
- Questionnaire designers must make a series of **trade-offs**, given the information needs of decision makers and the characteristics of target participants.
- We describe the objectives of a questionnaire and the steps involved in designing one, with guidelines for each step.
- The considerations involved in designing questionnaires for **international** marketing research are evaluated.
- The chapter concludes by examining how **digital developments** can help design more engaging questionnaires.

Real Research Example 1

Real research: Digital versus traditional newspapers: establishing reliability and validity

Survey Sampling International (SSI) is a global provider of sampling solutions – online and telephone, both fixed/landline and wireless/mobile, multimode and postal mail. A key part of its operations is the management of **access panels**. There has been much criticism of sampling quality issues addressed at the use of access panels. Online panel companies have done much to tackle quality issues and conduct research necessary to understand online panellists' motivation, behaviour, needs and wants.

Real research: (continued)

As panel providers, SSI researchers note: they see the questionnaires for themselves and receive feedback on them from panellists (some of whom 'vote with their feet' and drop out of the survey, or worse, leave the panel). At the same time as they strive to develop algorithms to identify poor quality data, they must endeavour not to penalise otherwise good panellists. There are real **'political'** and practical issues impacting the ability to make changes to questionnaire design.

Real Research Example 2

Real research: Brands in songs

The insertion of brands in songs is one of the most ignored forms of product placement. Researchers analysed the memorisation of 17 brands cited in two French songs ('Tes Parents' by Vincent Delerm and 'Wonderba' by MC Solaar), evaluating perceptions of placement and whether approval of a song and the artist generated a more favourable attitude towards the use of brands.

Real research: (continued)

Participants listened to two songs then were questioned about the brands mentioned, across three successive web pages: (1) general/music-interest questions, (2) knowledge, appreciation and spontaneous memorisation, (3) assisted memorisation and brand placement perception. The multi-screen design prevented participants from revising earlier answers and kept the study's true purpose hidden until after listening – average completion time was around 15 minutes.

Questionnaire Definition

Questionnaire

A structured technique for data collection consisting of a series of questions, written or verbal, that a participant answers.

Survey and observation are the two techniques for obtaining quantitative primary data in descriptive research. Both require a standardised procedure so data collected by different interviewers/observers are internally consistent, comparable and can be analysed uniformly.

A questionnaire is typically only one element of a data-collection package that might also include:

1. fieldwork procedures (selecting, approaching, questioning participants);
2. rewards, gifts or payment offered to participants; and
3. communication aids – maps, pictures, music, advertisements, products, return envelopes.

Three Objectives of a Questionnaire

Any questionnaire has three specific objectives:

1. Translate the information needed into a set of specific questions that participants **can** and **will** answer. Two similar ways of posing a question may yield different information – the most challenging objective.
2. **Uplift, motivate and encourage** the participant to become involved, cooperate and complete the task (see Figure 1).
3. **Minimise response error** – error that arises when participants give inaccurate answers, or when answers are mis-recorded or mis-analysed.

Figure 1 – Exchange of Values

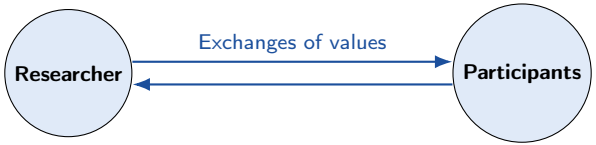
Figure 1. Exchange of values between researchers and participants

What the participant may want from the researcher:

- Tangible reward
- Confidentiality
- Interesting subject and experience
- Personal benefits from seeing research completed
- Social benefits from seeing research completed
- Being 'chosen' as an expert participant
- Research organisation known for excellence
- Rapport and trust

What the researcher may want from the participant:

- Honesty
- Takes in reasons for the study
- Follows instructions in completing the study
- Thinks through issues before answering
- Social benefits from seeing research completed
- Says good things about the rationale for research
- Says good things about the research process



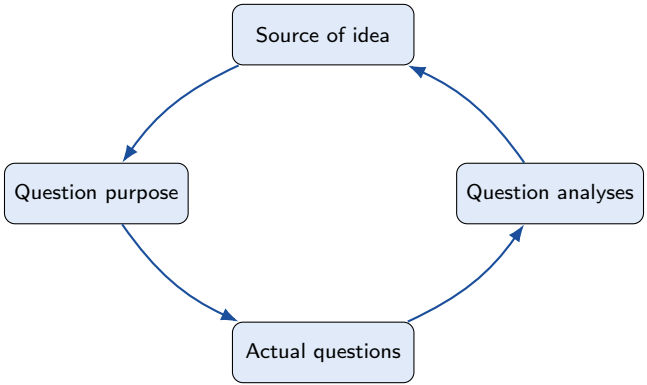
Questionnaire Design Process

The great weakness of questionnaire design is a **lack of theory**. Because there are no scientific principles that guarantee an optimal questionnaire, questionnaire design is a skill acquired through experience.

- The correct grammatical use of language does not guarantee the optimal questionnaire.
- Guidelines and rules help avoid major mistakes, but 'fine tuning' comes from the creativity of a skilled researcher.
- Developing the craft requires the creative **trade-off** of many factors (Figure 2).

Figure 2 – Trade-offs Faced by the Designer

Figure 2. Trade-offs faced by the questionnaire designer



The Design Process: Five Stages

The design process is founded upon generating information that will effectively support marketing decision makers:

1. The **'source of idea'** represents the culmination of decision makers' and researchers' diagnoses and the information available when a project is commissioned.
2. From the diagnoses emerge specific research questions – **'question purpose'** is established for each. Some research questions may not be addressed via questionnaires at all.
3. With clear question purposes, the process of establishing **'actual questions'** begins – researchers must put themselves 'in the shoes' of the potential participant.
4. **'Question analyses'** – how the data will be analysed – must be thought through from an early stage, not after questionnaires are returned. More sophisticated scale types (nominal $\rightarrow$ ordinal $\rightarrow$ interval $\rightarrow$ ratio) permit more powerful statistics, but are more onerous for participants.
5. The understanding taken from the data comes back to the **'source of idea'** – the cycle repeats as new data, interpretations or market forces emerge.

Real Research: Tango Facebook Questionnaire

Real research: Tango Facebook application questionnaire

When someone takes part in an intercept survey on a brand's site or Facebook Fan Page, the research becomes part of the brand experience. The questionnaire's look, feel and tone can all impact the brand relationship. A Tango Facebook application (created with Hall and Partners, Nudge and BBH) opened with:

"Bongiorno! So, we both know you've used the Tango Head Masher 3000 facebook app to mash more heads than a Zombie DJ at a funeral themed rave..."

Example question phrasing included: *'It made me feel that I could marry a can of Tango'* and *'It made me feel Tango was pretty bloody good.'*

Real Research: Trading Off Task with Statistical Power

Real research: Trading-off the participant task with statistical power

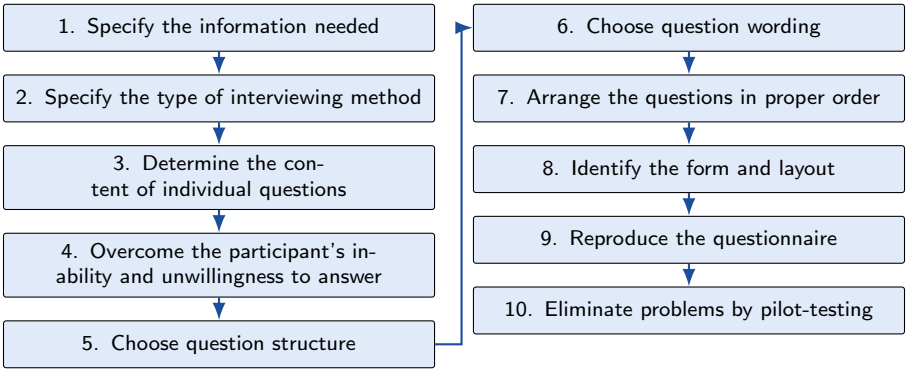
In a study of banking practices in Europe, participants were asked who they thought were the top four banks in their country (**ordinal** scale), then why the top two were perceived best. This could be done several ways:

- Tick matching characteristics from a list $\rightarrow$ easy, **nominal** data.
- Rank-order the same characteristics $\rightarrow$ more effort, more powerful **ordinal** data.
- Allocate 100 points using a constant-sum scale $\rightarrow$ most onerous, but powerful **interval** data.

The designer must weigh task burden against the meaning decision makers get from the findings.

Figure 3 – The Questionnaire Design Process

Figure 3. Questionnaire design process



Specify the Information Needed

This is the first step in questionnaire design – and also the first step of the research design process. As the project progresses, the information needed can become more clearly defined.

- Review the components of the problem and approach, particularly research questions, hypotheses and characteristics that influence research design.
- Prepare a set of **dummy tables** – blank tables used to present findings, portraying how the analysis will be structured once data are collected.
- Have a clear idea of the **target participants**. Wording/style appropriate for finance directors may not suit retired persons or schoolchildren.
- The more diversified the participant group, the more difficult it is to design a single questionnaire appropriate for the entire group.

Specify the Type of Interviewing Method

How the questionnaire is administered under each method shapes its design:

- **Online / postal** surveys are self-administered – questions must be simple, with detailed instructions.
- **Online surveys, CAPI, CATI** can easily accommodate complex skip patterns and randomisation of questions to eliminate order bias.
- **Face-to-face** interviews: participants see the questionnaire and interact with the interviewer – lengthy, complex, varied questions are possible.
- **Telephone** interviews: participants interact with the interviewer but do not see the questionnaire – limits questions to short and simple ones. Face-to-face and telephone questionnaires should be written in a conversational style.

Real Research: Effect of Interviewing Method (1/2)

Real research: Online and postal questionnaire design

"Please rank-order the following film genres in order of your preference. Begin with your most-preferred genre (1); continue until your least-preferred genre is ranked 10. No two genres share a rank. Just try to be consistent."

	Film genre	Rank order
1	Action	_____
2	Animated	_____
3	Comedy	_____
4	Drama	_____
5	Factual	_____

	Film genre	Rank order
6	Fantasy	_____
7	Light drama	_____
8	Romance	_____
9	Sci-fi	_____
10	Suspense	_____

Real research: Telephone questionnaire design

"I will read you the names of film genres. Please rate them on a scale of 1 to 10: 10 represents 'greatly preferred' and 1 represents 'not so preferred'... (READ ONE GENRE AT A TIME.)"

Real Research: Effect of Interviewing Method (2/2)

Telephone scale, 1 (Not so preferred) to 10 (Greatly preferred): rows for Action, Animated, Comedy, Drama, Factual, Fantasy, Light drama, Romance, Sci-fi, Suspense.

Face-to-face questionnaire design

(HAND GENRE NAME CARDS TO THE PARTICIPANT.) Please examine these cards carefully. Now give me the card with the film genre you prefer most. (RECORD THE GENRE AND TAKE THE CARD.) Would you carry on through the cards and hand over, in order, your second, third, fourth, etc.? (REPEAT SEQUENTIALLY UNTIL ONE CARD IS LEFT.)

- Ranking 10 genres is easy online/postal/mobile, but may be too complex over the telephone or email – a simpler one-at-a-time rating task is preferred there.
- Interviewer instructions (capital letters) are far more extensive for face-to-face interviews.
- Online/postal: the *participant* records ranks; face-to-face: the *interviewer* records them.

Is the Question Necessary?

Every question should contribute to the information needed or serve some specific purpose. If there is no explicit and satisfactory use for the resulting data, the question should be eliminated.

- Neutral questions at the start can establish involvement and rapport, particularly on sensitive/controversial topics.
- **Filler questions** disguise the purpose or sponsorship of the project (e.g. a Samsung-sponsored mobile-device survey may include Apple/Google filler questions).
- Questions unrelated to the immediate problem may be included to generate client support, or duplicated to assess **reliability** or **validity**.

Are Several Questions Needed Instead of One?

Once a question is necessary, ensure it is sufficient to obtain the required information unambiguously.

Double-barrelled question

A single question that attempts to cover two issues. Such questions can be confusing to respondents and result in ambiguous responses.

'Do you think Coca-Cola is a tasty and refreshing soft drink?' – a 'no' answer is ambiguous (not tasty? not refreshing? neither?). Two distinct questions are needed instead.

Multiple questions can also be embedded in a single **'why'** question (e.g. 'Why do you like the Odeon cinema?'), which may tap many different constructs at once. Most 'why' questions tap two aspects: (1) attributes of the product and (2) influences leading to knowledge of it – often better asked as two separate questions.

Overcoming the Participant's Inability and Unwillingness to Answer

Researchers should not assume participants can provide accurate, well-reasoned answers to all questions (assuming they are even willing to!). Certain factors limit participants' **ability** to provide the desired information – they may not be informed, may not remember, or may be unable to articulate certain responses.

Is the Participant Informed?

Participants are often asked about topics on which they are not informed. Marketers may have deep knowledge of their industries; target consumers may have little inclination to understand such intricacies.

Filter question

An initial question in a questionnaire that screens potential participants to ensure they meet the requirements of the sample.

- Filter questions measuring familiarity, product use and past experience should be asked before questions about the topic itself.
- Online, CATI and CAPI surveys allow extensive filtering, tailoring the questionnaire to familiarity and experience.
- A **'don't know'** option can reduce uninformed responses without reducing overall response rate – use it when participants may not be adequately informed.

Real Research: Corporate Social Responsibility

Real research: Understanding corporate social responsibility

In a Global Change Network survey among communications professionals, more than two-thirds believed consumers would be increasingly interested in social and environmental efforts – not at the expense of economic concerns, but in concert with them. Products demonstrating clear value *and* being good for society and the environment were likely to prosper. Consumers are often uninformed or confused by much of the news they get, and do not necessarily understand green terminology or how a company or product is really 'giving back'.

Can the Participant Remember?

Many things we might expect everyone to know are remembered by only a few. Evidence indicates consumers are particularly poor at remembering quantities of products consumed – reported usage has been found to exceed actual usage by 100% or more.

Better wording example

How often do you consume soft drinks in a typical week?

◯ Less than once a week ◯ 1 to 3 times a week ◯ 4 to 6 times a week ◯ 7 or more times a week

The inability to remember leads to errors of **omission**, **telescoping** and **creation**.

Telescoping

A psychological phenomenon where an individual telescopes or compresses time, remembering an event as occurring more recently than it actually occurred.

Aided vs. Unaided Recall

The ability to remember an event is influenced by: (1) the event itself, (2) the time elapsed since the event, and (3) the presence or absence of memory cues. We remember important, unusual or frequent events better; more recent events are remembered better.

Example – beer advertisement recall:

- **Unaided**: 'What brands of beer do you remember being advertised last night at the cinema?'
- **Aided**: list beer brands and ask 'Which of these brands was advertised last night at the cinema?'

In presenting cues, the researcher must guard against biasing responses by testing several successive levels of stimulation, then analysing the influence of stimulation to select an appropriate level.

Is the Participant Able to Articulate?

Participants may be unable to articulate certain types of responses – e.g. describing the 'atmosphere' of a cinema they would prefer. If provided with alternative descriptions, they can indicate the one they like best.

- If unable to articulate, participants are likely to ignore the question and refuse to respond to the rest of the questionnaire.
- Participants should be given aids – pictures, maps, descriptions – to help articulate their responses.
- Even when able to answer, participants may be **unwilling** because too much effort is required, the context seems inappropriate, no legitimate purpose is apparent, or the information is too sensitive.

Effort, Context and Legitimate Purpose

Effort required of participants. Most participants are unwilling to devote much effort. E.g. instead of asking participants to *list all items* purchased on their last shopping trip, provide a list of shops and ask them to indicate the applicable ones – less effort required.

Context. Some questions seem appropriate in some contexts but not others (e.g. personal hygiene questions fit a health-organisation survey better than a breakfast-cereal one). The context can sometimes be manipulated – e.g. "As a fast-food restaurant, we are very concerned about providing a clean and hygienic environment... we would like to ask some questions related to personal hygiene."

Legitimate purpose. Participants are unwilling to divulge information they don't see as serving a legitimate purpose. Explaining *why* data are needed ("To determine how preferences for cereal brands vary among people of different ages, we need information on...") increases willingness.

Sensitive Information

Participants may be unwilling to accurately disclose sensitive information – it may embarrass them or threaten prestige/self-image, or be seen as too personal/an invasion of privacy. If pressed, they may give biased responses, especially in personal interviews.

- Sensitive topics include money, personal hygiene, family life, political and religious beliefs, involvement in accidents or crimes.
- In industrial surveys, sensitive questions may cover much of what a company does, especially strategic activities and plans.

Increasing the Willingness of Participants

Participants may be encouraged to provide unwilling information via these techniques:

1. Place sensitive topics at the **end** of the questionnaire – once rapport and legitimacy are established.
2. **Preface** with a counter-biasing statement that the behaviour is common (e.g. "Recent studies show most European consumers are in debt.").
3. Ask using the **third-person technique** – phrase as if referring to other people.
4. **Hide** the question in a group of questions participants are willing to answer.
5. Provide **response categories** rather than asking for specific figures (e.g. income bands, not exact income).
6. Use **randomised techniques**: participants randomly answer one of two questions (one sensitive, one neutral with known probability), yes/no, without revealing which was answered – the researcher can estimate the population proportion via probability, but not identify individuals.

Choose Question Structure

A question may be **unstructured** or **structured**. We discuss their advantages/disadvantages, then the major types of structured questions: multiple choice, dichotomous and scales.

Unstructured questions

Open-ended questions that participants answer in their own words (also 'free-response'/'free-answer' questions).

e.g. 'What is your occupation?' / 'What do you think of people who patronise secondhand clothes shops?' / 'Who is your favourite film personality?'

Unstructured Questions – Pros & Cons

Advantages

- Good as first questions on a topic, or as a final 'catch-all' question.
- Much less biasing influence on response than structured questions – participants are free to express any view, offering rich insights.

Disadvantages

- Coding responses is costly and time-consuming (though software helps).
- Not very suitable for self-administered questionnaires (online, CAPI, email, postal) – participants tend to write more briefly than they'd speak.
- Extra weight implicitly given to more articulate participants.

Pre-coding can overcome some disadvantages: expected responses are recorded in multiple-choice format although presented to participants as open-ended (interviewer selects the matching category).

Real Research: Mobile vs. Fixed Phones for Surveys

Real research: Comparing mobile with fixed phones for surveys

A study of Portuguese adults (15+) compared mobile with fixed-telephone survey methods, run by the same CATI centre over the same period with the same interviewers on both. The questionnaire had eight nominal-scale questions, three batteries of ordinal scales (25 items), one open-ended quantitative question (hours online per week) and a demographics section. In the mobile sample, 50% of participants spent less than 5 hours a week online; in the fixed sample the same proportion spent less than 6 hours. In both samples 25% spent more than 14 hours; no significant difference at $p < 0.05$ – similar Internet usage intensity (nearly 10 hours/week on average).

Structured Questions – Multiple Choice

Structured questions

Questions that pre-specify the set of response alternatives and the response format: multiple choice, dichotomous, or a scale.

Multiple-choice example

Do you intend to buy a new watch within the next six months?

◯ Definitely will not buy ◯ Probably will not buy ◯ Undecided
◯ Probably will buy ◯ Definitely will buy Other (please specify) _____

The response alternatives should include the set of all possible choices, including 'other (please specify)', and should be mutually exclusive (unless multiple selection is explicitly allowed).

Order Bias

Order bias (position bias)

A participant's tendency to choose an alternative merely because it occupies a certain position, or is listed in a certain order.

- Participants tend to tick the first or last statement in a list, particularly the first.
- For a list of numbers (quantities/prices), there is a bias toward the central value.
- Online/electronic forms: response alternatives can be positioned **randomly** for each participant.
- Manually prepared forms: use different versions where alternatives vary from form to form, each appearing once in each extreme position, once in the middle, once in between.

Multiple-choice questions overcome many disadvantages of open-ended questions: quick administration, reduced interviewer bias, improved cooperation, and much less costly coding/processing.

Real Research: IKEA – Love Where You Live

Real research: IKEA – love where you live

Launching a new IKEA store in Perth, research showed the main barriers for new customers were: (1) the old store too far/long journey; (2) parking/ease of shopping; (3) IKEA not having products suited to particular tastes. A new, much bigger store instigated a communications campaign aimed at changing perceptions of IKEA – not just launching a store. An online survey (28 multiple-choice questions) on a purpose-built microsite, promoted via banners, in-store, direct mail and a radio tie-in, generated over 30,600 microsite visits, averaging 8 minutes on site, and 9,474 completed surveys against an initial target of 1,000.

Structured Questions – Dichotomous

Dichotomous question

A structured question with only two response alternatives, such as yes and no, often supplemented by a neutral alternative.

Dichotomous example

Do you intend to buy a new watch within the next six months?

◯ Yes ◯ No ◯ Don't know

- If a substantial proportion of participants can be expected to be neutral, include a neutral alternative; if the proportion is expected to be small, avoid it.
- Easiest to code/analyse, but the response can be influenced by the wording (**split-ballot technique**: frame the statement one way on half the questionnaires, the opposite way on the other half).

Structured Questions – Scales

Scales specify response alternatives along a continuum. Using the watch-purchase example as a 5-point scale:

Definitely will not buy	Probably will not buy	Undecided	Probably will buy	Definitely will buy
○	○	○	○	○

This is only one of several scales that could be used to ask this question (see Chapter 12 for more examples: Likert, semantic differential, constant-sum, etc.).

Choose Question Wording

Question wording is the translation of desired content and structure into words participants can clearly and easily understand – perhaps the most critical and difficult task in developing a questionnaire.

- Poor wording $\rightarrow$ participants may refuse to answer, or answer incorrectly (**item non-response** / **response error**).
- Unless participant and researcher assign the same meaning to the question, results will be seriously biased.

Eight guidelines:

1. Define the issue 2. Use ordinary words 3. Use unambiguous words
2. Avoid leading/biasing questions 5. Avoid implicit alternatives
3. Avoid implicit assumptions 7. Avoid generalisations/estimates 8. Use positive and negative statements

1. Define the Issue

Questions should clearly define the issue in terms of who, what, when, where, why and way (the **six Ws**).

Poorly defined

Which brand of shampoo do you use?

Ambiguous: does 'who' mean the participant personally or the household? What if more than one brand is used – most preferred, most often, most recent, or first that comes to mind? Is 'where' implied to be at home?

Better wording

Which brand, or brands, of shampoo have you personally used at home during the last month? In the case of more than one brand, please list all brands that apply.

2. Use Ordinary Words

Words should match the vocabulary level of target participants and/or be supplemented with graphic/visual support. Most participants do not understand technical marketing words.

- Instead of *'Do you think the distribution of soft drinks is adequate?'*, ask *'Do you think soft drinks are easily available when you want to buy them?'*
- Your language communicates and puts participants in a frame of mind as they answer – language that is not meaningful to them may put them in an unintended frame of mind.

Real Research: The Forgotten 12 Million

Real research: The forgotten 12 million

There are 12 million adults in the UK whose first language is English but who have a reading and writing age between 9 and 14 – almost a third of the adult population. These people can read and write, they are not illiterate, and may not be interested in improving their skill levels or acknowledge the issue publicly. Implications: they will not take part in research requiring reading/writing, will not volunteer that they have literacy issues, and are unlikely to take part if a questionnaire looks complicated. Questionnaires/forms may be completed by a partner or friend instead – their true views go unrepresented. Researchers must consider how to create a participation-conducive environment and stimulus materials suited to their needs.

3. Use Unambiguous Words

Words that seem unambiguous can have different meanings for different people – e.g. ‘usually’, ‘normally’, ‘frequently’, ‘often’, ‘regularly’, ‘occasionally’, ‘sometimes’.

Ambiguous

In a typical month, how often do you visit a boutique?

◯ Never ◯ Occasionally ◯ Sometimes ◯ Often ◯ Regularly

Better – objective categories

In a typical month, how often do you visit a boutique?

◯ Less than once ◯ 1 or 2 times ◯ 3 or 4 times ◯ More than 4 times

All-inclusive/all-exclusive words (‘all’, ‘always’, ‘any’, ‘anybody’, ‘ever’, ‘every’) should also be avoided – ‘any’ could mean ‘every’, ‘some’, or ‘only one’.

Choosing Words: Seven Checks

Consult a dictionary and thesaurus, and ask:

1. Does the word mean what we intend?
2. Does the word mean the same to our target participants?
3. Does the word have any other meanings?
4. If so, does the context make the intended meaning clear?
5. Does the word have more than one pronunciation?
6. Is there any word or similar pronunciation that might be confused with this word?
7. Is a simpler word or phrase suggested that may be more meaningful to our target participants?

4. Avoid Leading or Biasing Questions

Leading question

A question that clues the participant to the desired answer, or leads the participant to answer in a certain way.

This tendency to agree is called **acquiescence bias** (yea-saying).

Leading

Do you think that patriotic French people should buy imported cars when that would put French workers out of employment?
◯ Yes ◯ No ◯ Don't know

Better

Do you think that French people should buy imported cars?
◯ Yes ◯ No ◯ Don't know

Bias from Sponsor Mentions

Bias may also arise when participants are given clues about the sponsor of the project – they tend to respond favourably towards the sponsor.

- *'Is Colgate your favourite toothpaste?'* could bias responses in favour of Colgate.
- Better/unbiased: *'What is your favourite toothpaste brand?'*
- Mentioning a prestigious (or non-prestigious) name can also bias the response, as in: *'Do you agree with the British Dental Association that Colgate is effective in preventing cavities?'*
- Unbiased alternative: *'Is Colgate effective in preventing cavities?'*

5. Avoid Implicit Alternatives

Implicit alternative

An alternative that is not explicitly expressed in the options.

Making an implied alternative explicit may increase the percentage selecting it:

1. *Do you like to fly when travelling short distances?* (driving only implicit)
2. *Do you like to fly when travelling short distances, or would you rather drive?* (driving explicit)

The first question is likely to yield a greater preference for flying than the second. Questions with implicit alternatives should be avoided unless there are specific reasons for including them.

6. Avoid Implicit Assumptions

Implicit assumptions

Assumptions that are not explicitly stated in the question.

1. *Are you in favour of a balanced national budget?*
2. *Are you in favour of a balanced national budget if it would result in an increase in personal income tax?*

Implicit in Question 1 are consequences such as a cut in defence expenditure, an increase in personal income tax, a cut in health spending, and so on. Question 2 makes an assumption explicit. Question 1's failure to make assumptions explicit would result in overestimating support for a balanced budget.

7. Avoid Generalisations and Estimates

Questions should be worded so participants do not have to make generalisations or compute estimates.

Requires computation

What is the annual per capita expenditure on clothing in your household?

Participants would have to determine annual expenditure (multiply monthly by 12, or weekly by 52), then divide by household size – most are unwilling or unable to do this.

Better – two simple questions

What is the monthly (or weekly) expenditure on clothing in your household?
and
How many members are there in your household?

The researcher can then perform the necessary calculations.

8. Use Positive and Negative Statements

Many questions, particularly those measuring attitudes and lifestyles, are presented as statements with which participants indicate agreement/disagreement. Evidence shows the response obtained is influenced by the **directionality** of statements.

- It is better to use **dual statements** – some positive, some negative.
- Two versions of a questionnaire could be prepared: one half-negative/half-positive statements interspersed one way, the direction reversed in the other version.
- (An example of dual statements was provided in the summated Likert scale on attitudes towards the Odeon cinema, Chapter 12.)

Arrange the Questions in Proper Order

The order of questions is as important as the wording used. Questions communicate and set participants in a particular frame of mind – set at the start of questioning, and changed as each question is posed and answered. Designers must understand the **logical connections** between questions as perceived by target participants.

Opening Questions

Opening questions can be crucial in gaining participants' confidence and cooperation – they should be interesting, simple and non-threatening.

- Questions asking for opinions can be good openers, because most people like to express their opinions.
- Classification issues (age, gender, income) as openers tend to make participants concerned about the purpose of the whole survey, giving it the feel of an 'official form' (census, tax form) rather than a positive engagement.
- Where necessary to qualify participants for eligibility, **qualifying questions** serve as the openers (may be classification questions such as age).

Type of Information

The type of information obtained may be classified as:

1. **Basic information** – relates directly to the research problem.
2. **Classification information** – socio-economic/demographic characteristics used to classify participants, understand results and validate the sample.
3. **Identification information** – name, postal address, email, telephone – used to verify that listed participants were interviewed and to send promised incentives/prizes.

General guideline: basic information first, then classification, then identification – the basic information is of greatest importance and should be obtained before risking alienating participants with personal questions.

Difficult Questions & Effect on Subsequent Questions

Difficult questions – sensitive, embarrassing, complex or dull – should be placed **late** in the sequence, after rapport has been established and participants are involved.

Effect on subsequent questions. Questions asked early can influence responses to later ones. As a rule of thumb, general questions should precede specific questions (prevents specific questions from biasing responses to general ones).

Example

Q1: What considerations are important to you in selecting a boutique?
Q2: In selecting a boutique, how important is convenience of its location?

Funnel approach

A strategy for ordering questions where the sequence starts with general questions, followed by progressively specific questions, to prevent specific questions from biasing general ones. (The *inverted* funnel starts specific and ends general – useful when participants have no strong/formed views.)

Logical Order & Branching Questions

Questions should be asked in a **logical order** – as the target participants perceive logic, not necessarily the researcher. All questions on a topic should be asked before switching topics; brief transitional phrases help.

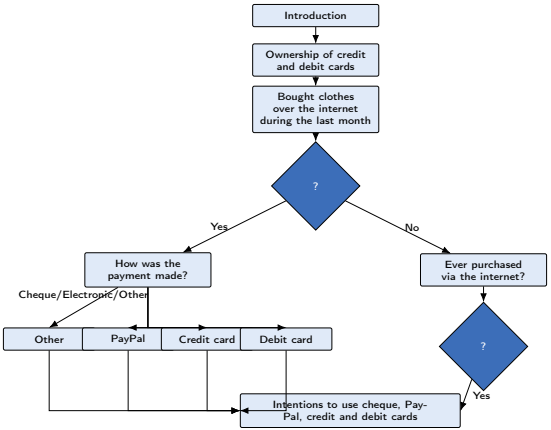
Branching question

A question used to guide a participant/interviewer through a survey by directing them to different spots depending on the answers given.

- Branching questions ensure all contingencies are covered, reduce interviewer/participant error, and encourage complete responses. A flow chart of logical possibilities helps design them (Figure 4).
- Guideline 1: the branched-to question should be placed as close as possible to the question causing the branching.
- Guideline 2: branching questions should be ordered so participants cannot anticipate what additional information will be required (otherwise they may discover they can avoid detailed follow-ups by giving certain answers).

Figure 4 – Flow Chart for Questionnaire Design

Figure 4. Flow chart for questionnaire design



Identify the Form and Layout

The format, spacing and positioning of questions can significantly affect results, particularly in self-administered questionnaires.

- Divide a questionnaire into several parts (e.g. basic information section); number questions in each part – eases coding, especially with branching questions.
- **Pre-coding**: codes to enter into the computer are printed on the questionnaire (built into the software for online/CATI/CAPI surveys).
- Hard-copy questionnaires should be numbered serially – controls questionnaires in the field, eases coding/analysis, helps track losses. (Exception: postal questionnaires – numbering may reduce participation if seen as identifying respondents.)
- Online questionnaires should not be designed with a 'paper survey' mindset – a flat, text-based experience can feel dull/monotonous next to rich, interactive websites people are used to (the 'Pong' analogy).

Reproduce the Questionnaire

How a questionnaire is reproduced can influence results (relevant to online, SMS and CAPI too, in terms of form/layout/tailoring).

- Poor-quality/shabby paper suggests the project is unimportant, adversely affecting response quality – use good-quality paper and a professional appearance.
- Multi-page printed questionnaires should take the form of a **booklet** rather than loose/clipped/stapled sheets – easier to handle, less likely to come apart, allows double-page spreads, looks more professional.
- Avoid splitting a question (including its response categories) across pages – can mislead into thinking the question has ended.
- Use **vertical** response columns (easier to read than sideways/split formatting); avoid crowding questions together just to look shorter – crowding causes data-collection errors and lower cooperation/completion.
- Place instructions as close as possible to the relevant question; use colour-coding for branching questions where helpful.

Eliminate Problems by Pilot-Testing

Pilot-testing

Testing the questionnaire on a small sample of participants for the purpose of improving the questionnaire by identifying and eliminating potential problems.

A questionnaire should not be used in the field without adequate pilot-testing. All aspects should be tested: question content, wording, sequence, form and layout, question difficulty and instructions.

Real Research: Re-inventing Long-Haul Travel

Real research: Re-inventing long-haul travel

Air New Zealand worked with Synovate to research all areas of the flying experience: seats, in-flight entertainment, food and beverage, service flow, kitchen design – also surfacing ideas not previously considered, such as on-board events. Attitudinal statements were combined from findings by design consultancy IDEO and Air New Zealand's own prior research. An online pilot questionnaire was tested on 40 long-haul travellers, clearly informed it was a pilot. Participants paused after each block of statements to reflect on clarity and whether statements captured the wider need. Open-ended responses and actual ratings (checked for discriminative power) were used to refine statements. The final questionnaire included 50 attitudinal statements across the whole long-haul experience.

Pilot-Testing – Practical Guidelines

- Participants in the pilot test should be similar to those in the actual survey (same population, background characteristics, familiarity, attitudes/behaviours).
- Pilot-tests are best done face-to-face, even if the actual survey will use online/postal/telephone methods – interviewers can observe reactions and attitudes. Further pilots by the actual method should follow.
- A variety of interviewers should be used – experienced interviewers detect uneasiness/confusion/resistance; novice interviewers help identify interviewer-related problems.
- Sample size typically **15 to 30** participants for initial testing (more if the target population is highly heterogeneous, or if piloting involves several stages/waves).
- **Protocol analysis**: participant 'thinks aloud' while answering; remarks are recorded and analysed. **Debriefing**: after completion, participants are told it was a pilot and asked to explain the meaning of each question, their answers, and any problems.
- **Editing**: correcting problems identified; sound pilot-testing involves several stages – one pilot test is a bare minimum.
- Analyse pilot-test responses against the pre-planned **dummy tables** – if a table remains empty, a necessary question may have been omitted.

Questionnaire Design Checklist (1/2)

Step 1: Specify the information needed

1. Ensure information fully addresses the components of the problem.
2. Prepare a set of dummy tables.
3. Have a clear idea of participant characteristics/motivations.

Step 2: Specify the interviewing method

1. Review the type of interviewing method to be used.

Step 3: Determine content of individual questions

1. Is the question necessary?
2. Are several questions needed instead of one?
3. Do not use double-barrelled questions.

Step 4: Overcome inability/unwillingness to answer

1. Is the participant informed?
2. Use filter questions if not likely informed.
3. Can the participant remember?
4. Avoid errors of omission, telescoping, creation.
5. Questions without cues can underestimate occurrence.
6. Can the participant articulate?
7. Minimise effort required.
8. Is the context appropriate?
9. Make the request seem legitimate.
10. If sensitive: (a) place at end; (b) preface with common-behaviour statement; (c) use third-person; (d) hide among willing-to-answer questions; (e) use response categories; (f) use randomised techniques if appropriate.

Questionnaire Design Checklist (2/2)

Step 5: Choose question structure

1. Open-ended questions useful for exploratory/closing.
2. Use structured questions whenever possible.
3. Multiple-choice alternatives: all possible choices, mutually exclusive.
4. Dichotomous: include neutral alternative if substantial proportion expected neutral.
5. Consider split-ballot technique to reduce order bias.
6. If numerous alternatives, use more than one question.

Step 6: Choose question wording

1. Define the issue in terms of who/what/when/where.
2. Use ordinary words matching participant vocabulary.
3. Avoid ambiguous words (usually, often, regularly...).
4. Avoid leading or biasing questions.
5. Avoid implicit alternatives.
6. Avoid implicit assumptions.
7. Avoid generalisations/estimates.
8. Use positive and negative statements.

Step 7: Arrange questions in proper order

1. Opening questions: interesting, simple, non-threatening.
2. Qualifying questions should serve as openers.
3. Basic, then classification, then identification info.
4. Difficult/sensitive/complex questions placed late.
5. General questions should precede specific questions.
6. Ask questions in logical order.
7. Branching questions designed to cover all contingencies.
8. Branched question close to the branching question; order so participants cannot anticipate required info.

Step 8: Design the form and layout

1. Divide into several parts; number questions.
2. Print coding on forms for manual data entry.

Step 9: Publish the questionnaire – visually engaging; vertical columns; grids for related questions; avoid crowding; instructions near questions; booklet format for long/hard-copy surveys. **Step 10: Eliminate problems by pilot-testing** – always test; small sample (15–30); vary interviewers; use protocol analysis/debriefing; analyse pilot data against dummy tables; don't aggregate pilot data with the final survey.

Designing Surveys Across Cultures and Countries

Questionnaire design should be adapted to specific cultural environments, avoiding bias in terms of any one culture, at every step of the design process.

- Balance information needs against differences in consumer behaviour, decision processes, psychographics, lifestyles and demographics.
- Classification variables (marital status, education, household size, occupation, income, dwelling unit) may not be directly comparable across countries – e.g. household definition/size varies greatly with extended-family structures.
- Survey methods that dominate in Western countries (e.g. online) may be less prevalent elsewhere – questionnaires may need to suit multiple modes, facilitated by online research communities.
- Adaptation must also address **comprehension and translation** – prefer two or more simple questions over one complex question.
- Consider variability in how informed participants are across cultures; unstructured/open-ended questions reduce imposed response alternatives but are more affected by differing education/literacy levels – use with caution in low-literacy contexts.
- Translated questionnaires require **linguistic equivalence**, verified via pilot-testing: on monolingual subjects in their native language *and* on bilingual subjects comparing original vs. translated versions, analysing response patterns for cultural bias.

Real Research: Online Research Communities

Real research: A cross-cultural review of participants' perceptions

Online research communities are the fastest-growing sector of marketing research, credited with 'authentic voice of the consumer' and 'increased engagement' – benefits a study tested by comparing views across five countries (Australia, Canada, China, Indonesia, USA) and survey modes (face-to-face, telephone, online panels, focus groups, online research communities).

Real research: (continued)

A distinction emerged between more-developed research markets (Australia, Canada, USA) and less-developed ones (Indonesia, China), with China part-way between. The two less-developed markets showed higher participation in online panels/other modes – a highly researched minority. Chinese participants were least positive about face-to-face research; Indonesians most positive about telephone research. Online research communities need to be more engaging or risk disengaging participants – improving incentives, recognition, feedback and information provided in return for participants' effort.